GUESS-ESTIMATES
A Primer

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GUESSTIMATES – A Primer

What?

Simply put, a Guesstimate = Guesswork + Estimate. Managers often have to take business decisions without adequate information. For this one has to arrive at a rough estimate of a metric based on educated guesses and calculation.

Why?

- Interviews - Recruiters are increasingly using guesstimates in the initial rounds of their selection process
- Business Decisions - Managers rely heavily on guesstimate techniques to support their business decisions
- Consulting - Guesstimates are used extensively in the field of consulting and almost all consulting interviews have a customary guesstimate round

Key Points for a Guesstimate Interview:

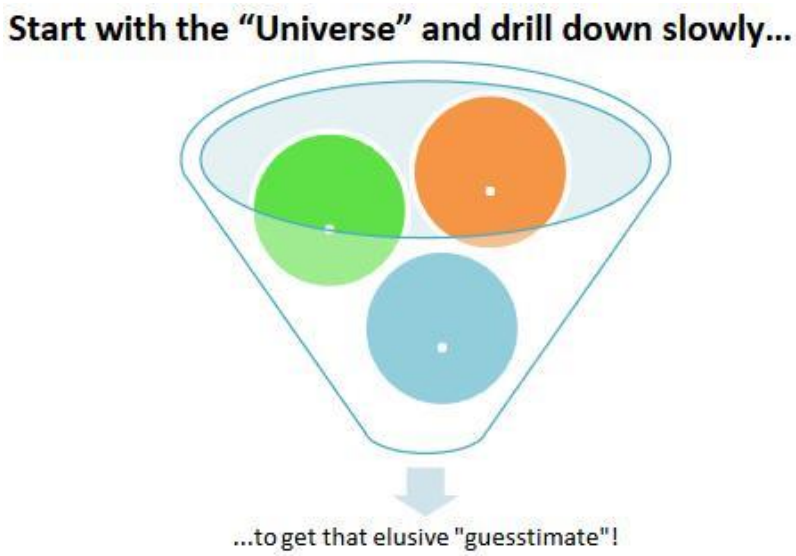
- Keep interacting with the interviewer and think aloud
- The approach to the answer is more important than the answer itself
- Give reasons for your guesses and assumptions

So, you walk into the interview room, exchange greetings and introduce yourself. You talk about your work and your hobbies and there you go, you get your first guesstimate. What follows below is an example of an actual guesstimate presented during summer processes.

Often, they are so off the wall to shake up an otherwise calm and collected candidate. In our previous experiences, a guesstimate is often related to your field of work or hobbies. The candidate is expected to keep calm, analyse the situation and conclude while thinking out loud.

Remember, it is a guesstimate, no one is expecting the candidate to arrive at an exact figure. The consultants in the room are more eager to see how you are able to think with an open mind, engage as many variables as possible and come towards a solution quickly eliminating excessive analysis. The main objective of these questions is *to evaluate your poise and professionalism* when facing an outlandish situation. How you react to this question when presented will speak volumes about your ability to be professional when faced with a similar business

situation during a client engagement. The best approach for a guesstimate or a brainteaser question is to think of a funnel. You begin by thinking broadly, then slowly drill down towards the answer.



Once you get your question, politely confirm the same with your interviewer. Take a minute to compose your thoughts and jot down points neatly onto the notepad. Once you feel you are ready to go, start involving the interviewer. Start with explaining your approach run him through all the factors you have already listed down. These factors should contribute to the final answer and also be clear to provide your assumptions for the same. Take the interviewers' approval and clarify the questions he poses. Once done, you may start assigning values, figures for each and arrive at the final solution.

Key success criteria will be to show how broadly you can think in terms of number of factors and to arrive at a solution at the end making appropriate assumptions.

Let us look at a few illustrations to get a further perspective –

How?

The target metric is usually spread over different categories. Split the categories into various sub-categories: up to a level where you can make an informed guess about the individual contribution of each of the sub-categories towards the target metric. Combine the contributions of all the categories to get to the final number.

Overall there are three main type of approaches to solve a guesstimate:

- **Consumption side/Demand side**
- **Production side/Supply side**
- **Random approach**

1. Consumption side

Here the question is usually solved using end consumer's data. For instance, for calculating number of cars in a city it would be better to start from population of the city and then following a top to bottom approach. Below are some examples of where and how this approach is used:

Estimate the total consumption of water every day within the MDI campus

You may start by segregating the consumption into various categories such as individuals, families, mess & eateries maintenance-related, etc.
Then split the categories, say individuals, into sub-categories such as residents and visitors. You may split residents further into students and support staff to get a better idea of the numbers in each sub-category.
Then make an informed guess about the consumption of each individual, say 5 litres for drinking, 30 litres for bathing, etc.; multiply this by the total number of individuals (say 600 students and 50 support staff) and you'll get the consumption of individuals. Similarly, make an estimate about the other categories and add them up.

Can you tell us the number of basketball in the city of Gurgaon right now?

Here it seems easier to use the **The Start Big Approach.** So it is easier to look at various avenues where basketballs could be used and then drill down to each variable and assign values to it. This is how the candidate ended up solving the question. Please note there is no right or wrong way to go about it.
Major avenues for basketball –

- Basketballs owned by people for playing with basketball (Basketball or any other game (dodgeball etc.)
- Basketballs owned by schools/colleges in its sports department
- Basketballs stored in shops
- Basketballs stored in factories in Gurgaon as inventory before shipping to shops

Based upon discussions with the interviewer, the candidate was asked to drill down on first two avenues - Basketballs owned by people for playing –

Total population of Gurgaon - 15 Lakhs					
Age group	**Population (lakhs)**	**% of people who play basketball**	**People who actually, own a basketball**	**No of Basketball (lakhs)**	**Comments**
Age 0 - Age 5	1.5	1%	80%	0.012	May be used as a toy in the house
Age 6 - Age 15	3	10%	20%	0.06	Used for basketball and dodgeball
Age 16- Age 25	3	20%	20%	0.12	Recreational activities
Rest	7.5	2%	20%	0.03	

Basketball owned by schools/colleges in its sports dept.

No of schools in Gurgaon – 250 (a known fact)

Taking an average 5 balls a school, considering all types of schools, total no of basketballs – 750 No of Colleges in Gurgaon – 100

Taking an average 10 balls a college, considering all types of colleges, total no of basketballs – 1000

One could have made it complicated by taking different types of school or colleges and then assigning a rate of number of balls etc., but as mentioned earlier, idea is to quickly arrive at a solution. So, we recommend mentioning the factors, but simplifying the steps at each stage taking the interviewers' approval.

2. Production side/Supply side

Production side or the supply side approach is better when making estimations using demand of a certain product or service is difficult for example estimating revenue of a restaurant in Cyber Hub, Gurgaon. In this case, we can assume the total capacity of the restaurant say 100 and then make estimations of the occupancy rate at any time of the day. To see the method in detail, below is an example:

- ***How many passengers fly out of Delhi in a day?***

Terminal	Total no of Departure	Avg capacity of a service	Avg occupancy	Total no of people
Domestic	400	160	80%	51200
International	200	300	85%	51000
			Total	**102200**

As you can see, here if we had started with Start big approach, say population etc., it would have been very difficult to drill down.

As a parting note on guesstimates, advice to candidates is to start with listing down as many factors/variables as possible, take interviewers comments and then drill down to quickly arrive at a solution. Some of usual factors include – population, city, demographic distributions, urbanization, income distribution, age, gender, time, season, etc.

3. Random approach

Here both demand and supply side approaches would not serve the purpose and you need to devise a random approach according to the question. For instance, calculating the number of trees in MDI Gurgaon or calculating weight of a big substance. Below is an example of the approach:

- ***How much does a 747 weigh?***

The first step is to clarify certain elements or just make assumptions (informing the interviewer about them) regarding certain peripheral factors:
Are there passengers on board? No.
Any baggage? No
Are the fuel tanks full or empty? Full.
Any food or beverages on board? No.

Now, just go ahead and calculate the weight of each part of the plane

- 8 full fuel tanks: I'll assume the plane can fly 6,000 miles and uses 10 gallons to the mile. So that's 60,000 gallons at 2 pounds a gallon equals 120,000 pounds.
- 18 tires: I'll assume that the tires weigh 200 pounds each — that's 3,600 pounds
- 4 engines: I'll assume 2,500 pounds each, which adds another 10,000 pounds
- 2 wings: 200 feet long by 30 feet wide is 6,000 square feet, times a square foot weight of 5 pounds times 2 wings equals 60,000 pounds.
- Interior: 75 rows of seats times 4 feet per row equals 300 feet. Add on the cockpit, bathrooms, etc. — let's say around 400 feet long. I assume that the average weight per foot is 10 pounds, which equals 4,000 pounds
- The seats: They number, say, 500 and weigh 10 pounds each, so that's 5,000 pounds.
- Air in the cabin: It's captured air so we need to add one ton for the air in the cabin — 2,000 pounds.
- The Aluminium exterior: It's pretty thin and lightweight. If the plane is 400 feet long by 25 feet high, then about 10,000 exterior square feet at 1 pound per foot equals 10,000 pounds.
- Miscellaneous materials: The tail, overhead bins, carpet, stairs, wiring, and bathroom fixtures add on, say, another 2,000 pounds.

1. Surf Excel Detergent Usage in India in a day?

India ~1.3 billion

Rural ~ 70% ~ 900 million

Urban ~ 30% ~ 400 million

Rural - upper BPL : ~ 40% = 360 million

Rural (BPL) : 60% 540 million

Rural (BPL)

We can assume Rural BPL mostly as collective households, averaging at 5 per family: ~110 million

Mostly they use sachets. let's average the usage 2 sachets of 3 rs each per week: 24–30 Rs. per month ~ 25*110 = 2750 million INR.

Now Surf Excel is a little costly sachet , and given cheaper alternatives available, we can assume its share to be around 5–10% =200 million

For per day: 200/30 ~ 7million INR

Rural (Above BPL)

We can assume 4 sachets per week ~ 12 = 50–60 per month

Total household: 72 million = 72*60 = 4320 million INR monthly

Surf Excel share ~ 20% (others going to wheel, rin, ok, ghadi etc.) = 864 million INR

Daily use therefore: 864/30 = 28.8 million INR

Total Rural daily revenue: ~ 35 Million INR

Urban

Urban Low class ~ 30% = 120 million = 22 million households: Usage 250gms per month: 50

INR = 1100 Million INR

Surf Excel share: 5–7% = 77 million INR.

Daily: 77/30 million INR = 2.10 million INR

Urban Middle class: - 40% = 160 million = 40 million households (4 per family)

Average use: 500 gms per month , 90 Rs per month = 3600 million INR

Surf excel share: 20–25% = 900 million INR

Daily : 900/30 = 30 million INR

Urban High Class: 120 million ~ 30 million households (4 per family)

Average use: 1kg per month, 200 INR = 6000 million INR.

Surf Excel (King here) ~ 70% share ~ 4200 million INR

Daily = 4200/30 = 140 million INR.

Total Urban daily revenue: = 172 million INR

So total daily monetary value of the usage:

Rural daily + Urban daily = 35+172 = ~ 210 million INR

So, Surf Excel totalling a value of Rs. 210 million is used in India daily.

If you want to find out the quantity, let's divide it by the average kg price ~ 160 INR = 210/160 Million Kgs ~ 1.3 Million Kilograms per day.

2. How many passengers travel daily in Delhi metro?

There are six metro lines in Delhi-

Assumptions to be made:

1. Average number of boxes in one metro = 7 (6 and 8) So number of Passengers in one metro= 7*150(in one box) = 1000(approx.)

2. Busy hours: 8 - 11 in morning and 5 - 10 in evening. Total = 8 hours

Others hours: 6 - 8 in morning, 11 - 5 in afternoon and 10- 11 in night Total=9 hours

Frequency of metro = 3 minutes (Busy hours), 6 minutes (Others hours)

Calculation Part:

For Busy hour Number of passengers:

Total metro through one route*Number of route*Number of Passengers in one Metro (Assume 1000)

= 2(both side) *8(number of hour) *(60/3) (frequency)*6*1000= 1920000

For non-busy hour in same way, Number of Passengers= 2*9*10*6*1000= 1080000

Total Passengers= 1920000+1080000= 3000000 (3 million)

3. What is the average number of people at any given time in the Mumbai airport on a weekday?

People at any point in time can be either those who are waiting at the airport for arrival of the flight or for the luggage to come after departure of a flight.

There are 7airlines and 15 flights suppose on an average per airline making total no. of flights at any point of time =15*7=105.

Frequency of departure is half an hour, let's say, this means when a flight arrives, some people arrive from the flight to the airport waiting for the baggage and some other passengers at that time are waiting at the boarding point. At the same point

in time, we have both types of passengers waiting in airport. Each flight has a capacity of 350 people suppose. So, the total no. of waiting passengers will be 105*(350+350) =70k(approx.). Also, we have 10–20 people (4–5 at the boarding counter,4–5
at the security counter+4–5 at the boarding point out gate) of the airline staff making it 20*7=140(max).
Cleaning staff of airport, security personnel at the airport and maintenance technicians would add to another 100. Also, you have restaurants/bars within airport where people would be serving food. This would be around 50.
I am not taking people waiting at the bar or having food in restaurant here separately as that has already been considered.
Total=70k+140+50=70.3k(total)

4. How would you estimate the number of weddings in India in a year?

In case where people are involved always start with the country's population.
India's Current Population = 1.285 Billion
Assume that people are linearly distributed from 1 year to 80 years old. People usually tend to get married at 20-30 years of age. I have taken this window targeting the rural landscape as well.
So, no. of people in 20-30 years of age will be = (10/80) * 1.285 Billion =160 million
But as we already know that India has comparably large youth population and there is a tiny
fraction that gets married beyond 20-30 years of age. I would increase this figure to 180 million.
Since marriage is dependent on the female population and India professes to have an
average sex ratio of 900 in 1000 males.
Calculating by this approach (0.9/ (1+0.9)) *180 million = 85.2 million female population that gets married
I have to assume that each person gets married only once and then after reaching the appropriate answer introduce and error of 2% in the calculation since few people India go for second marriages.

Also, it will be safe to assume a linear distribution in age people get married since it is greater than 26 in urban areas and less than 26 in rural areas.
So, the total marriages will be 85.2 / 10 = 8.52 million marriages. Now adding an approximate error of 2% as mentioned above = 8.7 million marriages in a year.

5. How would you estimate the number of fruits sold in India in a day?

Let's start by the current Indian population= 1.285 billion. From this figure, we have to eliminate the population which is unable to afford eating fruits.
According to the figure stated in Wikipedia, 23.6% people are below poverty line. I would approximate this figure to be 25% which comes out to be approx. 320 million.
So, now net population = 1285 million -320 million = 965 million
Let us categorize people into age groups and assuming a linear distribution of ages across people (average life expectancy 80 years)
1. 1-15 = 181 million ~ 200 million
2. 15-50 = 423 million ~ 465 million
3. 50-80 = 362 million ~ 300 million
Since our population is has a fair majority of youth population, I have shifted the figures appropriately.
Now, we have to assume the average consumption of fruits in these age groups.
1. 1-15 = 1 fruit per day = 200 million fruits
2. 15-50 = 2.5 fruits per day = 465 * 2.5 = 1162.5 million fruits
3. 50-80 = 1.5 fruits per day = 300 * 1.5 = 450 million fruits
Total fruits = 1812.5 million
Since this figure is totally based on individual consumption and does not focus on commercial aspects, it would be fair to multiply this figure by 1.2!
So, net consumption of fruits in a day= 2175 million

6. Estimate the number of office chairs sold in India?

Total Population of India = 1.28 billion
Working population = (Age 22 to 60 years) = 50% of 1.28 billion = 0.64 billion
We can split up the age group of 22 to 60 years in terms of income as follows,
Below Poverty line = 20%

Lower Middle Class = 40%

Middle Class = 30%

Upper Middle Class =10%

Let us assume that people in Below Poverty Line work in blue collar jobs which involve

frequent transit (Auto drivers, fruit vendors, security staff etc)

Let us assume that 30% of lower middle-class work as clerks/related jobs at offices and all of

middle and upper middle-class work at offices which are stationary (desk work requiring chairs)

Relevant population = (30%*40%*0.64) + (40%*0.64) = 0.25 billion

Assuming a sex ratio of 1:1, let us assume that 50% of women and 80% of men work (This takes care of unemployment)

Total working population using chairs = (0.5*0.125) + (0.8*0.125) = 0.1625 billion = 160+ million

OR

The current population of India (as of July 01,2015) - 1.28 billion India's Population 2015

Let us subcategories the population into Rural and Urban.

Assuming an even distribution of number of people in a particular age group and considering the average life expectancy to be 80 years.

No. of people in the 21 to 60 age group will be = 0.5 * 1.28 billion = 0.64 billion

Roughly India's 30 % of population lives in urban areas and the rest 70% in rural areas.

Urban population = 30% of 0.64 billion = 192 million

Rural population = 448 million

Urban Population -

Now assuming that from the total of the workable class only 30% is educated/has a job/has

business/ has something to do with sitting in an office = 30% of 192 million = 57.6 million

Rural Population-

Since most of the rural class is related to the farming / labour intensive activities, I would

assume that only 10% of the people use office chairs in administrative / government offices
etc. = 10% of 448 million = 44.8 million
Total = 57.6 + 44.8 = 102.2 million

7. How many total Gmail users are there in India?

Let's start with the population of the country: ~ 1.3 billion
Rural: >70% ~ 870 million Urban: 30% = 430 million.
Rural Internet penetration ~ 10–15% (15%) = ~130 million
Urban internet penetration: 70% ~ 300 million
Also, Internet penetration in India overall: ~35% = 455 million (almost same as 130+300=430)
Now, let's talk about rural.
Rural user's majority uses internet for entertainment, some for Facebook. We can say very few
for emails.
So, 30% have email accounts. = 39 million
But still today Gmail is the most prevalent free email provider, so we can assume 80% market
share, rest goes to yahoo!, Hotmail, Gmail! (yes, Gmail is leveraging the letters of Gmail: D)
So rural Gmail users: ~32 million
Now Urban:
We can safely assume 95% have Gmail accounts. ~ 285 million
The rest 5% can be for the first-time internet users, newly on board/urban low class with very
rare access to internet, and other email users.
So total Gmail users in India: 285+32 = 317 million.

8. Guesstimate for the number of laptops sold in Delhi on an average routine day?

The Population of Delhi = 10 Mn.
Age band:

0–18: 25% = 2.5 Mn

19–50: 45% = 4.5 Mn

Rest = 3 Mn

Now people of Age group 0–18 won't be buying a personal laptop and might use their parents

or will have a desktop.

Age group 19–40, College people and Professionals will all be having one laptop at least.

There is no reason of taking in the Gender Bias over here. So, we can remove that.

Rest won't be buying it again for the reason mentioned for Age Group 0–18.

PC Penetration Delhi = 20 % (India = 10%)

People having a laptop or a desktop = 10% (City people usually have their own)

People having a laptop = 12% (Since majority of the people will be office goers)

So, the total would be = 4Mn * 0.12 = 480,000 People have a laptop

People buy a laptop at an average of 2 years, (company laptops are sturdier making them last

long)

Laptop sold per year = 0.5 * 480,000= 240,000

Laptops sold per day = 240,000 / 365 = 657

We can say at the best guess; 650 laptops are sold daily in Delhi.

9. How do I estimate number of ambulances on the road?

Let's try to solve this for Urban India.

Let's start with the population of the country ~ 1.3 Billion (1300 million)

Rural - 70% = 900 million Urban 400 million

Let's divide the Urban population into three groups based on income level - Low |High

|Upper High

I would divide as: Low: 30% High - 50% Upper High - 20%

So Urban Low: 120 million High - 200 Upper High - 80 million

Now out of Urban Low I would assume 10% have driving as occupation (Rise of taxi services

etc. (only considering 4-wheeler drivers))

= 12 million (120,00,000)

Out of these I would assume 1–1.5% are ambulance drivers = 120000. So, we can say almost 1,20,000 ambulances are available in the country at any time.

Now generally ambulances are available on calling, that means, at least a half of them are always on backup. let's assume that 70% of them are on idle at any given point of time.

Also, since their average journey time is of 20–25 minutes, so we can say not more than 10% of ambulances would be on road at any time So we can say 10% are on roads at any time = 12000 ambulances on road across Urban India.

10. How can the number of (pairs of) shoes sold online in Delhi through Flipkart in one day be estimated?

Delhi Population = 18 million

Considering the income pyramid, bottom 20% fall below poverty line, so won't buy online, another 20% fall in the low-income group, so we can discount them as well.

Remaining 60% have the tendency to buy stuff online, so 60% of 18 million is 72 lakhs. Now internet penetration in India is 15%, on an average. But in Delhi, it is bound to be higher.

Considering that we have already removed the bottom 40% of the population, the internet using population would be around 50% of the 72-lakh people, which becomes 36 lakh people.

Of these 36-lakh people how many would be open to do online transactions? around 10% =

10% of 36 lakh = 3.6 lakh

of these 3.6 lakh, how many would need shoes? On an average, men buy 2 shoes every year

and women 4 (assuming)

So, Average = (4+2)/2 = 3 per person

so, these 3.6 lakh people would buy 21.6 lakh (3.6* 3) shoes a year.

In a single day how many shoes will be sold? -? 21.6lakh/365

Out of this number, what % of shopping will be done on Flipkart? = around 30%

= 30% * 21.6/365 = ~1700 shoes

11. What is a guesstimate of YouTube's revenue on the weekends?

Total Population = 1.2 billion

You Tube Users = 3%

Let's say on an average a You Tube user watches around 70 videos in a month.

Total Video Views (Month): 1,200,000,000 * 3% * 70

Assuming a 30-day month, Total Video Views (Daily): 1,200,000,000 * 3% * 70 / 30

Let's assume that on an average there is 20% increase in viewing on weekends

Total Video Views (Weekends): 1,200,000,000 * 3% * 70 *1.20 *2 / 30 = 201,600,000 views

(Multiplied by 2, considering Saturday and Sunday as weekends)

Let's assume that YouTube is able to Monetize 20% of the views.

Monetizable Views = 201,600,000 *0.20

On an average CPV is 0.45

YouTube earnings through monetizable views = 201,600,000 *0.20 * 0.45 = INR 18,144,000

YouTube charges around INR 600,000 per day for its Masthead ads

Total Earnings of you tube on a Weekend in India = INR (18,144,000 + 1,200,000) = INR 19,344,000

Out of the earnings from monetizable views, YouTube gives back 55% to the content creators (people who have uploaded the videos) and keeps 45% as platform owners.

Net Earnings = INR (18,144,000*0.45 + 1,200,000) = INR 9,364,800

12. How many t-shirts do you think e-commerce companies selling in India per day?

We can approach this problem in two ways 1) Demand side 2) supply side I am going to use demand side:

Total population of India: 1 bn (approx.)

Reach to internet: 40% =400 Mn

Reach of ecommerce companies to deliver product: 3/4th = 300Mn

Assume 50% are male and 50% are female

Male = 150 Mn Female = 150 Mn

Let's solve for male: Now we will divide on the basis of age group because demand for each age will be different for example college students will have more t shirt compare to corporate person & old person.

0–15 yr. = 45 Mn, on an average individual have 4 t shirt -> 4*45=180 Mn

16–22 yr. = 23 Mn, on an average individual have 4 t shirt -> 4*23 = 92 Mn

23–50 yr. = 65 Mn, on an average individual have 3 t shirt -> 3*65 = 195 Mn
50 - 80 yr. = 18 Mn, on an average individual have 2 t shirt -> 2*18 = 36 Mn
Total t shirts owe by men: 180 + 92+195+36 = 503 Mn ~ 500 Mn.
Let's solve for female: also, device them on the basis of age groups
0–15 yr. = 45 Mn, on an average individual have 2 t shirt -> 2*45=90 Mn
16–22 yr. = 23 Mn, on an average individual have 4 t shirt -> 4*23 = 92 Mn
23–30 yr. = 15 Mn, on an average individual have t shirt -> 3*15 = 45 Mn
30 - 80 yr. = 67 Mn -> we can neglect this section
Total t shirts owe by women: 90+92+45 = 227 Mn ~ 230 Mn
Total t shirts owe by men + women = 500+230=730Mn
Average life of a t shirt = 2 year
Demand per year = 365 Mn ~ 360 Mn
Online platform provides coupons and offers but because of trust factor and fitting issues, people still in India prefer to buy offline.so we are assuming 30% of people buy t shirt through ecommerce portal and 70% buy using shops & showrooms.
Sell through ecommerce platform = .3*360 = 108 Mn ~ 100 Mn per year per day = 100 * 10^6/365 ~ 27,000.

The "START BIG" approach & The "START SMALL" approach

13. How many gallons of white house paint are sold in the U.S. every year?

THE "START BIG" APPROACH: If you're not sure where to begin, start with the basic assumption that there are 270 million people in the U.S. (or 25 million businesses, depending on the question). If there are 270 million people in the United States, perhaps half of them live in houses (or 135 million people). The average family size is about three people, so there would be 45 million houses in the United States. Let's add another 10 percent to that for second houses and houses used for other purposes besides residential. So, there are about 50 million houses. If houses are painted every 10 years, on average (notice how we deftly make that number easy to work with), then there are 5 million houses painted every year. Assuming that one gallon of paint covers 100 square feet of wall, and that the average house has 2,000 square feet of wall to cover, then each house needs 20 gallons of paint. So, 100 million gallons of paint are sold per year (5 million houses x 20 gallons). (Note: If you want to be fancy, you can ask your interviewer whether you should include inner walls as well.) If 80 percent of all

houses are white, then 80 million gallons of white house paint are sold each year. (Don't forget that last step!)

THE "START SMALL" APPROACH: Take a town of 27,000 (about 1/10,000 of the population). If you use the same assumption that half the town lives in houses in groups of three, then there are 4,500 houses, plus another 10 percent, which makes 5,000 houses. Painted every 10 years, 500 houses are being painted in any given year. If each house has 2,000 square feet of wall, and each gallon of paint covers 100 square feet, then each house needs 20 gallons – so 10,000 gallons of house paint are sold each year in your typical town.
Perhaps 8,000 of those are white. Multiply by 10,000 you have 80 million gallons.
Reference:

http://www.bc.edu/clubs/bcfa/docs/vault/Brainteasers%20and%20Guesstimates.pdf

14. Estimate the market for bicycles India.

- India has a population of 1.2B; India has ~30% population living below the poverty line. It would be fair to assume that a negligible proportion of poor population owns bicycles.
- Of the remaining 0.84M, approximately 50% fall in the age group of 8-40, the core bicycle using population. This amounts to ~0.42M people.
- Of these, assuming equal distribution between sexes, there are ~0.21M males and ~0.21M females.
- Here we assume that 75% of men buy at least single cycle in their lifetime and at least 20% women buy single cycle at some point. That makes a total of ~0.2M.
- Let's divide the population further into high users, middle users and low-income groups. Assume high income is 10%, middle income is 60% and low income is 30% Assume that the high-income group buys a cycle every 5 year; middle usage buys a cycle every 10-year, low usage buys 1 during their lifetime
- Calculating, cycles bought in a year = 0.2 (0.1/5 + 0.6/10 + 0.3/30) = 0.2 (0.02 + 0.06 + 0.01) = ~0.02M
- Assume the average price per denim to be 1000Rs. Thus, market size = Rs. 20B

Answer: Market size of cycles bought in India is ~Rs. 20M

15. Estimate the litres of cola sold in Delhi per day

- Population of Delhi is approximately 15M people.
- Assume 30% of population is below poverty line, does not consume cola.
- Population above poverty line is approximately 10M people. Cola drinking population is typically in age group 5-40 years of age. Approximately 60% of Delhi's population is in this range, i.e. ~6M people.
- Dividing this population into High consumers (20%), Moderate consumers (40%) and Low consumers (40%). Assume high consumers consume approximately 1L per day, moderate consumers consume 0.5L per day and low consumers consume 0.1L per day.
- Thus, total daily consumption = 1.2*1 + 2.4*0.5 + 2.4*0.1 = 1.2 + 1.2 + 0.24 = ~2.6M litres per day.

Answer: ~2.6M litres of cola is consumed in a day in Delhi

16. Estimate the number of denims sold in India in a year.

India has a population of 1.2B; India has ~30% population living below the poverty line. It would be fair to assume that a negligible proportion of poor population wears denims.

Of the remaining 0.84M, approximately 50% fall in the age group of 6-35, the core denim wearing population. This amounts to ~0.42M people.

Of these, assuming equal distribution between sexes, there are ~0.21M males and ~0.21M females.

Here we assume that 80% of men buy at least single denim in a year and at least 50% women buy single denim.

Let's divide the population further into high income, middle income and low-income groups. Assume high income is 10%, middle income is 60% and low income is 30% Assume that the high-income group buys 3 denims per person per year; middle income buys 2 denims per person per year, low income buys 1 denim per person per year.

Calculating, male denims bought = 0.8*0.21*(0.1*3 + 0.6*2 + 0.3*1) = 0.8*0.21*0.18 = 0.03M; calculating female denims bought = 0.5*0.21*(0.1*3 + 0.6*2 + 0.3*1) = 0.02M. Thus, total = 0.05M

Assume the average price per denim to be 1000Rs. Averaging for branded products in the 1800Rs pus range and non-branded in the ~500Rs. range. Thus, market size = Rs. 5B

Answer: Market size of denims bought in India is ~Rs. 5B

17. Estimate the amount of petrol used in Delhi per day

- Population of Delhi is approximately 15M people.
- Assume family size of 5, there are approximately 3M households in Delhi. Approximately top 40% households would own a car. Thus, there are approximately 1.2M cars in Delhi.
- We can divide these car users into Heavy, Moderate and Light users. Let's assume heavy users are 20% of total car owners and travel approximately 80km a day, to and fro work in Delhi-NCR etc. Assume Moderates are 50% of total car owners and travel
  ~30 km per day. Assume Light users are 30% of population and travel 10km a day
  (including low usage multiple cars owned by families etc.)
- Thus, total distance traversed by Delhites in a day = 1.2*(0.3*80 + 0.5*30 + 0.3*10) =
  1.2*70 = 84M km.
- Assume average mileage of a petrol car is 12km/l. Thus, amount of fuel consumption in a day = 7M litres

Answer: ~7M litres of petrol is consumed in a day in Delhi

18. Estimate the market size of paint industry in India

Working assumption – The household sector contributes to approximately 70% of the paint industry in terms of volume of paint used. The industrial sector accounts for the remaining 30%. We will first calculate the contribution of the HH sector and use this as a proxy for industrial segment.

India has a population of 1.2B, India has 70% population living in rural India. It would be fair to assume that a negligible proportion of rural population uses paints.

Given an urban population of 36M and an assuming an average family size of 5, number of households in urban India are 7.2M

Assume that only 75% of urban houses use paint or related products. Some houses in

India still use lime (chuna) to paint walls. Thus, relevant households are 5.4M

Assuming an average wall space in every household to be 5000 sq. ft, the amount of wall space painted in Indian urban households is = 27*109

Further, one gallon of paint can be used to approximately paint 200 sq. ft area. Thus, the number of gallons required to paint these urban household walls is ~ 14*107 gallons of paint

Using this figure to represent 70% of the volume of paint used, we can estimate the remaining volume of paint used by industrial sector, this would amount to ~6*107 gallons of paint

Considering that a household gets painted once in every 2 years, annual volume of paint required = 10*107 gallons of paint

Considering industrial sector gets painted once in 4 years (average), annual volume of paint required = ~3*107 gallons of paint

Hence, the total annual consumption is approximately 13*107 gallons of paint across all segments.

Adding 10% to this for rural consumption and another 10% for painting of newly constructed households, we get a total consumption of 16*107 gallons of paint

Assuming an average price of Rs 1000 per gallon of paint, the market size of paint annually in India is approximately Rs. 16*109

Answer: Market size of paint in India is approximately Rupees 1600 crores or 1.6B Estimate the amount of water that can fill an aircraft

Working assumption – This estimation is done taking into consideration Boeing 747, the most common airplane in India. Also, the plane is considered to be empty from inside, i.e. there are no seats, baggage cabins installed.

Let us assume the length of the plane to be ~70m, internal width of the plane can be calculated taking into consideration the seat and gallery space, given a 3*3 configuration of seats, each set of 3 seats would approximately be 2.5m wide. Thus, internal cabin width is approximately 6m. Similarly, the height of cabin can be estimated – the cabin allows a 6ft. man to stand comfortably, cabin height is approximated at 7ft. or 2m

The cabin space represents a cuboid with rounded edges. Estimating the volume of the cuboid, we arrive at 840 cu.m

Correcting for loss of volume due to rounded edges, decrease this by 10%, arrive at cabin volume of ~760 cu.m

Assuming cockpit, food gallery, washroom areas cumulatively account for ~10% of cabin area, estimated volume of plane is ~830 cu.m

Assuming cargo space to be of dimension of 2m (height), 8m(width) and 10m (length), cargo space volume is ~160 cu.m
Adding up, approximately 1000cu.m of water can fit in a Boeing 747

Answer: ~1000cu. m of water can fit in a plane

19. Estimate the number of red lights in Delhi

Working assumption – Number of red lights refers to the traffic signals on the roads of Delhi, not any other lights which are red in colour!
We know that Delhi is approximately 1600 sq.km in area. This is pretty much evenly divided between North, South, East and West
Imagine Delhi to be a square area with dimensions 40*40 as length and breadth. We divide the entire city into blocks of equal area, hence, 20 blocks from east to West and 20 blocks from North to South (with each block representing 2 sq. km)
Every intersection of four adjacent blocks would give us a cross road. Each crossroad would have a traffic signal. Hence, the number of signals at crossroads would be 400 across Delhi, averaging out for different areas.
Since there are 400 crossroads in Delhi, the number of red light at each crossroad is 4.
Hence, the total number of red lights at crossroads in Delhi is 1600
Let us assume that there are 20% additional red lights to the ones found at crossroads that are meant to control traffic on straight roads. These would amount to ~300 red lights.
Answer: 1900 red lights in Delhi

20. Estimate the number of leather balls sold in India in a year

Working assumption – Number of leather balls primarily implies leather cricket balls.

India has a population of 1.2B. Assuming almost equal distribution between both sexes, India has 0.6B males. Here we assume that a very small fraction of girls play cricket and can be ignored for this calculation.
India has 50% population below the age of 25 years. Applying this ratio to male population, there are 0.3B men in India below 25 years of age.

Assuming equal distribution of people in the age 0-24, there are approximately 0.2B males in the age group 10-25.

According to the World Bank, ~50% of Indian population lives on less than US 2$ per day, or approximately less than 100Rs per day. Given the approximate cost of a cricket leather ball is 150-200 Rs, we can assume that a very small fraction of leather balls is bought by this segment.

This reduces our target set of potential customers to 0.1B males in the age group 10-25

In a cricket crazy nation like India, let us assume 75% of these have played cricket at some point in time. This gives us 75M cricket players in India.

Of these, let us assume 10% have played with leather balls and 5% have bought a leather ball ever in their lives. Thus, approximately 3.8M Indians have ever bought a leather ball.

Of these, we can further assume that 50% are one-time purchasers, 40% are sporadic purchasers and 10% are regular purchasers.

Of the one-time purchasers (~2M), distributing over 20 years of life at an average, yearly purchase is equal to 1,00,000.

Of the sporadic purchasers (1M), assume they buy a single leather ball every 2 year. Thus, every year they buy 5,00,000 leather balls.

Of the regular purchasers (0.8M), assume they buy 2 leather balls per year. Thus, they buy 16,00,000 leather balls yearly. (These include the professional players and team members of various cricket clubs – assume, at an average, the association/club buys 2 balls per payer per year)

Accounting for all kinds of purchasers, total leather balls bought in India in a year = 22,00,000

Adjustments:

Adding 5% of this number to account for balls used in national/international cricket matches = 1,10,000

Also, adding 2% for balls bought by those with income below $2 per day and for girl cricket enthusiasts = 44,000

Total = 22,00,000 + 1,10,000 + 44,000 = ~ 23,50,000

Answer: ~23,50,000 balls are bought every year in India

21. Estimate the number of one-rupee coins that would fit in a normal size room

Working assumption – The room is empty, i.e. bare, without any furniture/person/object.

Let's assume the dimensions of a normal size room to be as follows, Height = 3m; Length = 4m, Breadth = 5m Volume of the room = 60 m3
Consider a one-rupee coin to be a square of area 2cm * 2cm. (Since when two one-rupee coins are kept adjacent, the space in between will remain unused and effectively, the space taken up by each coin would equal the space taken up by a square of side equalling the diameter of the coin) o Let the thickness of the coin be 2mm = 0.2 cm = 0.002m o Volume of the coin = 0.02 * 0.02 * 0.002 = 0.0000008 m3
Number of coins that would fit into the room = volume of room/volume of one-rupee coin (incorporated as a square) = 60/0.0000008 = 7.5 * 107

Answer: Number of coins that would fit a normal size room are 7.5 * 107

22. Estimate the number of vehicles crossing the road in front of MDI.

Working assumption: A normal working day in DU north campus with all colleges open. People using vehicles are Students, Teachers, Staff, Residents and Visitors while types of vehicles are Cars, two wheelers, autos/ rickshaws, Buses, cycles
Estimating the traffic based on time:
o 0000 - 0600 hours: 5 vehicles per hour (2 cars, 2 two wheelers and 1 cycle/rickshaw/ auto) buses won't run at that time
Total vehicles: 30 o 0600 - 0900 hours: 10 vehicles per hour (3 cars, 3 two wheelers, 3 cycles/rickshaw/ auto and 1 bus)
Total vehicles: 30 o 0900 - 1800 hours: 100 vehicles per hour (30 cars, 30 two wheelers,
35 cycles/ rickshaw/ auto, 5 buses)
Total vehicles: 900 o 1800 - 2200 hours: 50 vehicles per hour (10 cars, 20 two wheelers, 15 cycles/ rickshaw/ auto, 5 buses)

Total vehicles: 200 o 2200 - 2400 hours: 10 vehicles per hour (5 cars, 5 rest) no buses.
Total vehicles: 20

Answer: Total number of vehicles crossing the road in front of MDI in the day: 1200 approx.

23. Estimate the number of number of subscribers of Economist in India

Population of India: 120 crores
Economist target market is urban upper middle class, educated
Percentage of population in target: 20% (24 crores)
Percentage of people interested in economics, intelligent articles etc.: 10% (2.4 crores)
Percentage of people who prefer reading hard copy magazines: 20% (50 lakhs approx.)
People choosing economist among similar magazines (time, Newsweek etc): 40% (20 lakh)
Hence estimated subscriptions will be less as there may be more than 1 person using a subscription
Total number of subscriptions: 15 lakhs
Add subscriptions by libraries, colleges, corporates etc: 1 lakh apex

Answer: Total subscriptions approx. 16 lakhs

24. Estimate the number of cars sold by Maruti in India.

Population of India= 120 crores
% of population willing to purchase cars= 30% of 120 crores= 36 crores
% of population looking for cars in higher range (above 7 lakhs) = 10% of 36 crores= 3.6 crores
Popular brands in this category =5
Maruti cars sold in higher range (assuming equal sale of all cars= 3.6/5=720000 cars

Life of a car= 10 years

Cars in higher range sold annually = 72000 cars

% of population looking for cars in lower range (below 7 lakhs) = 20% of 36 crores= 7.2 crores

Popular brands in this category = 8

Maruti cars sold in higher range (assuming equal sale of all cars= 7.2/8=900000 cars

Life of a car= 15 years

Cars in higher range sold annually = 60,000 cars

Answer: Total Maruti cars sold in India in a year= 60,000 + 72,000= 1,32,000

25. Estimate the number of beers sold in Goa every day.

Population of Goa: 15 lakhs (approx.)

Percentage of population drinking per day: 20% (3 lakhs)

Average amount of beer consumed per person: 3 pints

Total beer consumed by local: 9 lakh pints

Tourist population on an annual basis:20 lakh (approx.)

Tourist population on a day (assuming tourist season is on, December first week):50,000

Percentage of tourist population drinking: 90% (45000)

Per capita consumption: 5 pints

Total consumption by tourists: 2.25 lakhs pints

Answer: Total in a day during season = 23 lakh pints approx.

26. Estimate the market for air conditioners in Bangalore in 1 year.

Population of Bangalore: 80 lakhs approx.

Two types of ACs sold: Domestic use and Commercial use.

For the purpose of this guesstimate we will assume only domestic ac sales.

ACs can only be afforded by middle class and above

Percentage of population falling in above category: 40% (approx.) = 32lakhs

Average size of household: 4 Number of households: 8 lakhs
Bangalore has pleasant weather throughout the year hence people who can avoid spending the money will do so.
Also, fresh grads and students form a very large percentage of the population. They will also avoid buying ACs even though they fall in the target demographic.
Number of households thinking of buying AC = 60% of target = 5 lakhs approx.
Average age of an AC = 5 years
Average number of ACs in a household of 4 = 3
Number of target ACs = 15 Lakh

Answer: The number of sales of air conditioners in Bangalore in a year = 15/5 = 3 lakh.

Additional Examples

27. Estimate the number of working ATMs in NCR?
28. How many Maggi packets are used at Sharma Ji in a week?
29. Estimate weekly revenues of the Sharma ji (Assume normal week and no seasonal variations)
30. Estimate the number of males who get a haircut in Delhi in 1 day?
31. How many people in India will be in flight at any given point of the day?
32. What is the number of flights leaving the Mumbai terminal each day?
33. What is the market size of disposable diapers in India?
34. Estimate the total number of chairs in the MDI Campus
35. What is the total number of cars sold in Delhi per year?
36. Estimate total rail-line network in India
37. Estimate the number of vehicles crossing the road in front of MDI.
38. Estimate the number of popcorns sold in PVR Saket every day.
39. Estimate the market for Pepsi in India.
40. Estimate the number of number of subscribers of Economist in India.
41. Estimate the market for Iphone8 in India
42. Estimate the number of MacBook's sold in India per day
43. How many Android users are there in India?

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Useful Data Sources

http://www.census2011.co.in/states.php.

http://www.indiademographics.com/default.aspx